LECTURE 5

EARTH EMBEDDINGS AND LOCATION ENCODERS

Geospatial Representation Learning

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Learning Outcomes

Lecture

- Explain the concept of Earth embeddings as AI-centric representations of geographic locations.
- Distinguish between embedding databases and neural location encoders.
- Describe how location, scale, time, and context can be encoded in geospatial representation learning.
- Evaluate the strengths and limitations of explicit and implicit geospatial embedding approaches.

Lab

- Query or construct a small geospatial embedding database.
- Use location-based embeddings for similarity search, prediction, or spatial interpolation.
- Implement a simple location encoder or use an existing pretrained location representation.
- Compare location-based representations against conventional coordinate or hand-crafted features.

PRACTICAL 5

LOCATION ENCODERS AND EMBEDDING DATABASES

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